

Quantum reconnection of an inclined vortex ring with a straight vortex line: Energy distribution and geometric analysis

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We investigate the reconnection process involving a quantum vortex ring colliding at different angles with a quantum straight vortex line. The dynamics of this phenomenon is well described by the Gross-Pitaevskii equation. Two families of initial configurations are explored, one by varying the angle α between the vortex ring and the horizontal plane, the other by changing the angle σ between the vortex ring and the vertical plane through the straight vortex and the ring center. Results show that sound generation is maximal when the ring is initially parallel to the straight vortex, whereas the kinetic energy is predominantly composed of the incompressible contribution when the angle α is increased. We provide evidence for the existence of a critical angle $\bar{\alpha}$ above which a gradual decrease in kinetic energy accompanied by an increase in the curvature of the ring after the second reconnection appears. Since curvature is associated with the velocity of the nodal lines, we conclude that the loss of the fluid's incompressible kinetic energy is compensated by the increase in the kinetic energy of the nodal lines. We also observe that for angles greater than $\bar{\alpha}$, the kinetic and internal energy components exchange their rate, while the quantum energy increases at a constant rate regardless of the angle. Changing the angle σ may result in both reconnection and absorption of the vortex ring, the latter occurring only for a restricted range of negative angles σ . A critical negative value $\bar{\sigma}$ is found beyond which the growth rate of the kinetic and internal energy changes sign. This occurs precisely when reconnection is replaced by absorption. We also observe that while changing the sign of α ensures symmetric behavior, changing the sign of σ instead leads to an asymmetry due to the effect of the straight vortex on the ring motion. In particular, for negative values of σ , the fluid kinetic energy decreases because the straight vortex opposes the ring, with a stronger and more evident effect as σ moves to less negative values.

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I. INTRODUCTION

Numerous phenomena have been investigated in the context of quantum vortex reconnections. The asymmetry and irreversibility of the reconnection process are of particular importance [1–3] and are confirmed by works [4,5] that showed an increased rate of change in the total length of nodal lines after the reconnection event with respect to the preceding phase, as well as by a larger curvature of the vortex filaments after the reconnection [1]. The increase in curvature, according to the local induction approximation (LIA) [6], leads to an increase in the velocity of vortex defects and thus to a faster separation time compared to their approach.

Total energy, which can be seen as the sum of four components (quantum, kinetic, potential, and internal energy [7]), is

conserved throughout the quantum reconnection phenomena. In [8], the behavior of the components of the total energy during the reconnection dynamics of two antiparallel quantum vortices has been analyzed, observing that the kinetic and quantum energy are converted into potential and internal energy, followed by a local kinetic energy depletion that resembles the energy decay in a classical fluid, even though the governing equations are energy conserving.

The kinetic energy can be further decomposed into its incompressible and compressible contributions through the Helmholtz decomposition [9,10]. The incompressible component is related to the generation of Kelvin waves [11] and the compressible one is related to the emission of sound waves [2,5,12], which is interpreted as one of the phenomena responsible for the observed asymmetry and irreversibility of the reconnection process. An earlier work [13] studied three reconnection processes involving two antiparallel vortices, the motion of two vortices initially placed at a right angle, and a sequence of reconnections and concurrent acoustic emissions. For each dynamics, the exchange between quantum energy, compressible kinetic energy, and incompressible kinetic energy is studied and reveals that the propagation of the sound waves transforms the quantum energy due to the vortex cores to the incompressible kinetic energy.

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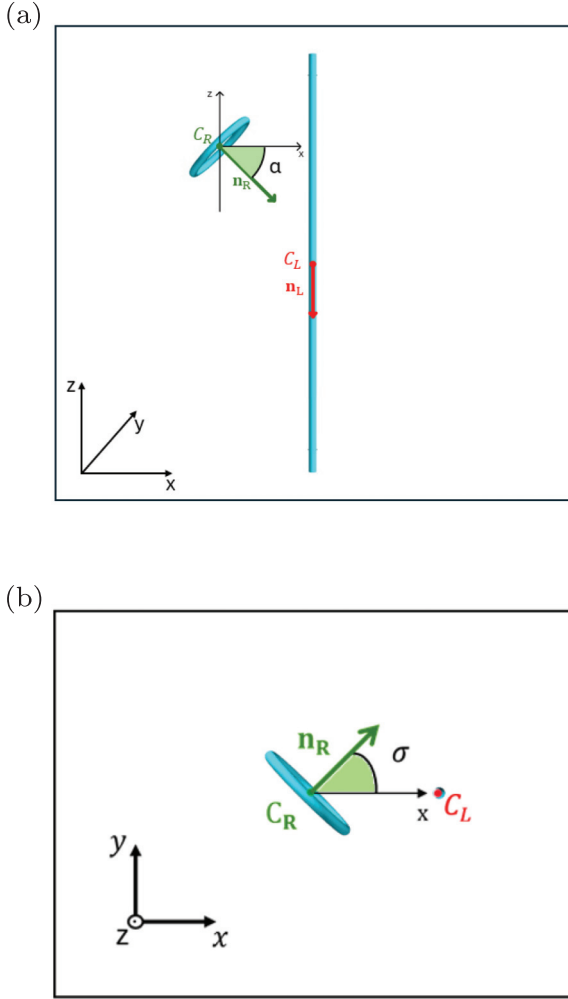


FIG. 1. Initial configuration for the two different settings, at $t = 0$, of the vortex ring and vortex line. C_R and $\vec{n}_R$ denote the center and the normal vector of the ring's plane, respectively. $\vec{n}_L$ indicates the direction of the vortex line, and $C_L = (0, 0, 0)$ represents the origin. (a) Initial configuration for the first setting. The angle α , defined in the xz plane between the axis x and the initial propagation direction of the vortex ring, is varied. (b) Initial configuration for the second setting. The angle α , defined between the initial propagation direction of the vortex ring and the vertical plane passing through the straight vortex line and the ring center, is varied.

In this work, we investigate the evolution of the reconnection between a straight vortex line and a vortex ring, whose dynamics is governed by the Gross-Pitaevskii equation (GPE) [14,15], considering two different initial configurations.

The first initial setup [see Fig. 1(a)] extends that in [16] and [17], by keeping the offset parameter fixed at zero and introducing a nonzero inclination angle α between the vortex ring and the horizontal plane, which is instead varied to explore its impact on the reconnection dynamics. In [16] and [17], the different choices in the initial geometrical setting lead to different reconnection dynamics with respect to the results in [16,17] that go beyond those obtained by varying the offset parameter at a fixed angle.

A setup more similar to ours is that considered in [18], where both the offset and the inclination angle are varied. However, the resulting dynamics may still differ since the present work employs the GPE model, which takes care naturally of the reconnection process, whereas the vortex filament method, which is used in [18], needs a reconnection strategy provided by a specific method.

The second initial configuration we consider [see Fig. 1(b)] is characterized by a varying angle σ between the vortex ring and the vertical plane. Unlike the cases considered in previous literature, this setup is largely unexplored. We expect that initial symmetric angular configurations will lead to asymmetric dynamics due to the different effect of the straight vortex in accelerating or decelerating the vortex ring.

For each angular value and for both configurations, we first compute the total length and curvature of the nodal lines, and then analyze how the components of the total energy redistribute during the process. Subsequently, by applying the Helmholtz decomposition to the kinetic energy following the approach of [13,17], we analyze how angle variations influence the partitioning between the incompressible and compressible components, revealing the generation of Kelvin waves and emission of sound waves. Finally, the results of the total and kinetic energy analysis are systematically compared with the measurements of vortex line length and curvature, aiming to identify possible correlations between the inclination angle of the vortex ring, geometric features, and energy redistribution during the reconnection. This analysis provides new insights into the role of the geometry of the initial configuration in quantum vortex reconnections and offers a framework for interpreting the energy transfer mechanism in quantum vortex dynamics.

II. PHYSICAL MODEL, GEOMETRIC TOOLS, AND ENERGY DECOMPOSITIONS

A. Physical model and numerical details

The dynamics of quantum vortices is successfully described by the GPE [7]. This is, indeed, an excellent model for studying the reconnection process between a quantum vortex ring and a straight quantum vortex line because reconnections occur naturally, without the need to force or model them [2]. The nondimensional form of the GPE reads [1,19]

$$\frac{\partial \psi}{\partial t} = \frac{i}{2} \nabla^2 \psi + \frac{i}{2} (1 - |\psi|^2) \psi, \quad (1)$$

where $\psi = \sqrt{\rho} e^{i\theta}$ is the complex wave function, which allows a fluid-dynamics interpretation in terms of fluid density $\rho = |\psi|^2$ and velocity field $\mathbf{u} = \nabla \theta$ via Madelung transformations [20]. Quantum vortices are phase defects characterized by a density profile that monotonically increases from zero at the vortex centerline to a uniform background value (normalized to 1) as $|x| \rightarrow \infty$. As such, they evolve over time within a medium of uniform density, except in the vortex core region. The size of the vortex core is of the order of the healing length ξ , which turns out to be equal to 1 due to the normalization employed while deriving (1). The (nondimensional) total energy of the system is constant under the GPE [1,7,14,15,21] and can be decomposed into

four contributions,

$$E_{\text{tot}} = \underbrace{\frac{1}{8} \int \frac{|\nabla \rho|^2}{\rho} d^3 \mathbf{r}}_{E_q} + \underbrace{\frac{1}{2} \int \rho |\mathbf{u}|^2 d^3 \mathbf{r}}_{E_k} - \underbrace{\frac{1}{2} \int \rho d^3 \mathbf{r}}_{E_p} + \underbrace{\frac{1}{4} \int \rho^2 d^3 \mathbf{r}}_{E_{\text{int}}}, \quad (2)$$

where E_q is the quantum energy, E_k is the kinetic energy, E_p is the potential energy, and E_{int} is the internal energy.

We investigate the reconnection phenomenon numerically using (1), with initial conditions given by an inclined vortex ring moving toward a steady straight vortex line. The circulation Γ of both vortices is 2π due to the normalization [7]. The initial wave function ψ_0 at time $t = 0$ is

$$\psi_0 = \sqrt{\rho_{0L}\rho_{0R}} \exp[i(\theta_{0L} + \theta_{0R})], \quad (3)$$

where ρ_{0L} and ρ_{0R} denote the density distribution of the straight vortex line and the vortex ring, respectively, while θ_{0L} and θ_{0R} are the corresponding phases. Both densities are computed using the fourth-order Padé approximation (see [22]), while θ_{0L} is set to change by 2π in a plane orthogonal to the vortex line; for the vortex ring, the phase θ_{0R} follows the construction described in [4].

For the first initial configuration at $t = 0$, the vortex line passes through the origin of the domain, which we call C_L [see Fig. 1(a)], and has direction $\vec{n}_L = (0, 0, -1)$. The vortex ring, with radius $R = 8$, is centered in $C_R = (-20, 0, 15)$ and has normal vector $\vec{n}_R = (1, 0, -\tan \alpha)$, where α is the angle in the xz plane between the axis x and the normal $\vec{n}_R$. In the second configuration, at $t = 0$, the vortex line and the vortex ring have the same initial condition of the first setting with the exception that the vortex ring, with radius $R = 8$ and centered in $C_R = (-20, 0, 15)$, has normal vector $\vec{n}_R = (1, \tan \sigma, 0)$, where σ is the angle in the xy plane between the axis x and the normal $\vec{n}_R$ [see Fig. 1(b)].

The spatial domain is $\Omega = [-40, 40] \times [-30, 30] \times [-55, 35]$ with spatial resolution $\Delta x = \Delta y = \Delta z = \xi/2$ (where ξ is the healing length), i.e., the number of points is $161 \times 121 \times 181$. The numerical integration of (1) is performed using second-order equispaced finite differences with homogeneous Neumann boundary conditions for the spatial discretization, whereas the time integration employs the second-order Strang splitting method as in [19,23]. The time step τ is set equal to $1/10$, while the final time T is the simulation time at which the vortex ring reaches the boundary of the computational domain. We remark that for each chosen value of the angles α and σ , the final time T is different; therefore, all the plots in Sec. III have different final times on the t -axis depending on the values of α and σ .

In general, when the vortex ring collides and reconnects with the straight vortex, numerical simulations show a dynamics that evolves through three different stages: initially, before reconnection, the system consists of two separate structures, the vortex ring and the straight vortex line; then the first reconnection takes place, and a single continuous curve is observed. Finally, a second reconnection occurs, giving rise to a (non-straight) vortex filament and a nonplanar vortex ring. Figure 2

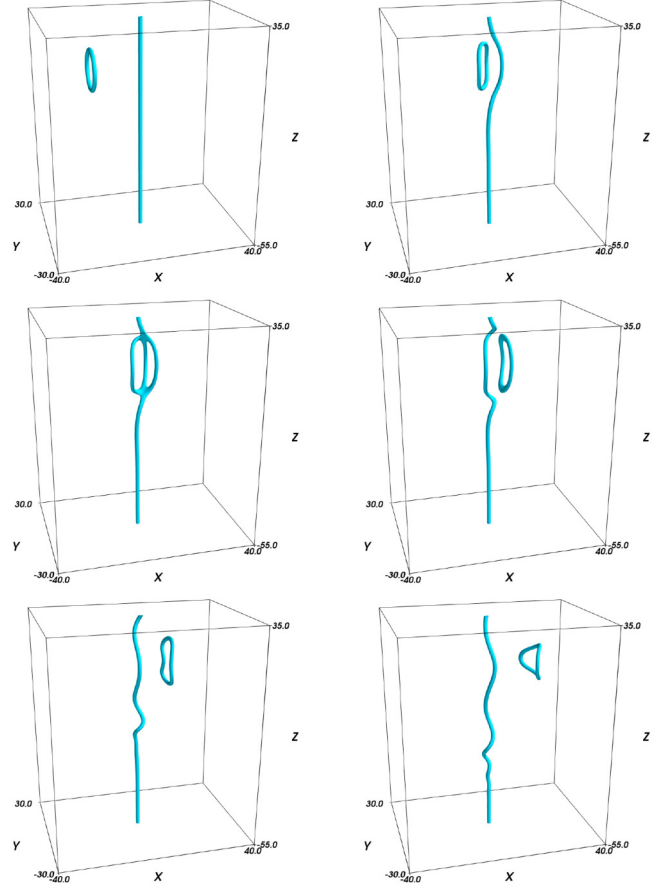


FIG. 2. Frames of time evolution of the reconnection process for initial angles $\alpha = \sigma = 0^\circ$. We can see the two reconnection moments in the second and fourth frames. Isosurfaces of $\rho = 0.2$.

reports the time evolution of the configuration $\alpha = \sigma = 0^\circ$. (For a complete set of simulations, see the Supplemental Material [24]). We note that loop solitons, observed for a similar setup in [18] where the vortex filament method is employed, are not present in our reconnection process, possibly because of the use of different models or slightly different geometrical parameters. Moreover, we do not observe any absorption of the ring by the vortex line in this setting $\alpha = \sigma = 0$, since the offset parameter is zero, but we expect to obtain it for $\sigma \neq 0$ in the study of the second configuration. The three-step pattern described above, which will recur throughout the work, suggests that the curvature and the length of the filaments would change quite considerably during the reconnection process and that Kelvin and sound waves would be generated. In what follows, we introduce the tools to track them.

B. Geometric tools

We analyze the evolution of the total length and curvature of the vortex-core nodal lines, numerically extrapolated as in [1], during the reconnection process, aiming to understand a possible relation between these geometric quantities. We recall the definitions of total length L and curvature K :

$$L := \sum_{i=1}^N \oint_{C_i} ds, \quad K_i := \frac{1}{L_i} \oint_{C_i} c_i(s) ds, \quad (4)$$

where $i = 1, \dots, N$ is the number of nodal lines $\mathcal{C}_i$, s is the arc length, L_i is the length of the single nodal line, and $c_i(s) = ||\mathcal{C}_i''(s)||$ is the local curvature. We underline the fact that we consider the total curvature of the nodal lines and not the local one.

C. Decomposition of kinetic energy

To understand the possible generation of Kelvin and sound waves, we consider the decomposition of the kinetic energy of the system into its incompressible and compressible contributions by applying the well-known Helmholtz decomposition to the vector field $\sqrt{\rho}\mathbf{u}$. By doing so, we have

$$E_k = E_k^i + E_k^c, \quad (5)$$

where $E_k^i = \frac{1}{2} \int ((\sqrt{\rho}\mathbf{u})^i)^2 d^3\mathbf{r}$ is the incompressible component, which characterizes the velocity field of the fluid around the vortex core and is related to Kelvin wave generation [11], and $E_k^c = \frac{1}{2} \int ((\sqrt{\rho}\mathbf{u})^c)^2 d^3\mathbf{r}$ is the compressible one, which is related to sound wave emission [3,5,12]. Helmholtz decomposition is indeed applied to the field $\sqrt{\rho}\mathbf{u}$, thus yielding $\sqrt{\rho}\mathbf{u} = (\sqrt{\rho}\mathbf{u})^i + (\sqrt{\rho}\mathbf{u})^c$, where the incompressible component $(\sqrt{\rho}\mathbf{u})^i$ satisfies $\nabla \cdot (\sqrt{\rho}\mathbf{u})^i = 0$, and the compressible component $(\sqrt{\rho}\mathbf{u})^c$ satisfies $\nabla \times (\sqrt{\rho}\mathbf{u})^c = 0$ (see, e.g., [13,25] for details). The decomposition of $\sqrt{\rho}\mathbf{u}$ is here computed using its scalar-potential formulation, i.e., by the Poisson equation,

$$\nabla \cdot (\sqrt{\rho}\mathbf{u}) = -\Delta\Phi,$$

with Φ a scalar potential function. We express the quantity $\sqrt{\rho}\mathbf{u}$ as $\text{Im}(\nabla\psi e^{-i\theta})$, where Im indicates the imaginary part, and θ is the phase arising from the Madelung transformation (see [20]). We numerically solve the Poisson equation using the Fourier method, and once Φ is obtained, the compressible and incompressible components are computed as $(\sqrt{\rho}\mathbf{u})^c = -\nabla\Phi$, $(\sqrt{\rho}\mathbf{u})^i = \sqrt{\rho}\mathbf{u} - (\sqrt{\rho}\mathbf{u})^c$.

III. NUMERICAL RESULTS AND THEIR ANALYSIS

In this section we report a detailed analysis of the numerical simulations carried out by changing both the initial angle α [between the normal plane to the vortex line and the normal to vortex ring, see Fig. 1(a)] and the angle σ [between the xz -plane and the normal vector $\mathbf{n}_R$ of the vortex ring, see Fig. 1(b)]. For the first configuration we have considered $\alpha = 0^\circ, \alpha = 15^\circ, \alpha = 30^\circ, \alpha = 45^\circ, \alpha = 60^\circ$ and $\alpha = 75^\circ$, as shown in Fig. 3, whereas for the second configuration we have first studied the qualitative behavior by varying the angle σ so as to check the range in which reconnections occur, and then performed a more detailed analysis for some specific cases in each range of σ . (All the numerical simulations are available in the Supplemental Material [24]). For the first configuration, we see that, regardless of the value of the angle α , both the vortex ring and line undergo deformations before the first and after the second reconnection. Additionally, the first reconnection occurs once the deformed vortex ring touches the deformed vortex line. We stress that we also considered the symmetric angles $\alpha \rightarrow -\alpha$, setting the center of the ring with an opposite z -coordinate and imposing the z -dimension of the spatial domain as the interval $[-35, 55]$,

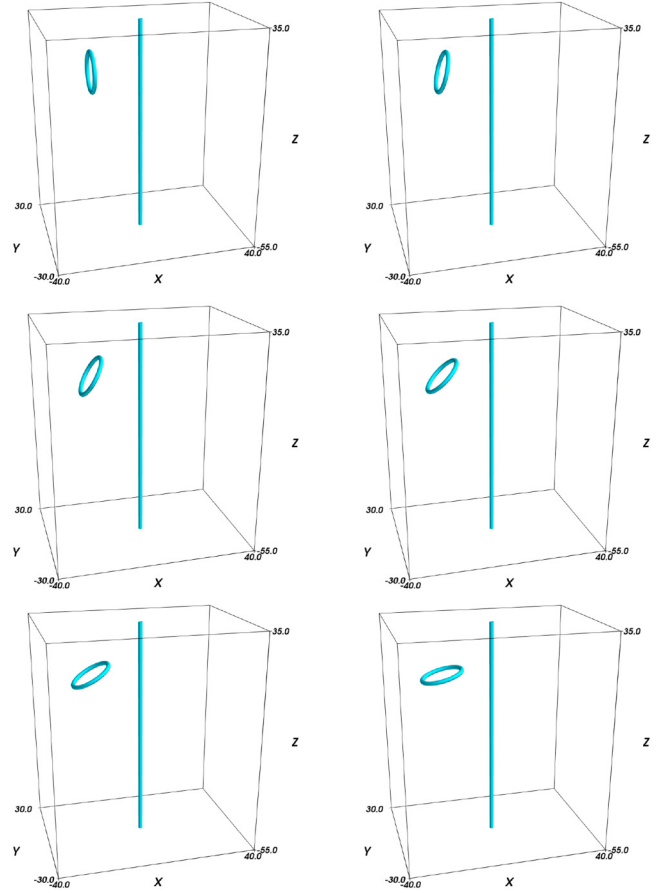


FIG. 3. Configurations of initial conditions with different values of angle α . From top left to bottom right, $\alpha = 0^\circ, \alpha = 15^\circ, \alpha = 30^\circ, \alpha = 45^\circ, \alpha = 60^\circ$, and $\alpha = 75^\circ$. Center of the vortex ring: $C_R = [-20 \ 0 \ 15]$. Ring normal coordinates: $\tilde{\mathbf{n}}_R = [1 \ 0 \ -\tan \alpha]$. Center of the vortex line: $C_L = [0 \ 0 \ 0]$. Line direction coordinates: $\tilde{\mathbf{n}}_L = [0 \ 0 \ -1]$. Isosurfaces of $\rho = 0.2$ are plotted to visualize the vortex cores.

in order to visualize the full dynamics of the reconnection. We do not report here the results for the negative values of α since, as foreseen, we observed a perfect symmetry in the evolution of the reconnection dynamics with respect to the xy -plane, as well as identical outcomes in the analysis of the total length, total curvatures, and in the decomposition of the total and kinetic energy. (For completeness, all the corresponding results are provided in the Supplemental Material [24]). This symmetry is clearly due to the role of the straight vortex line, whose circulation does not “asymmetrize” the dynamics, since the angle α is varied with respect to the xy -plane. On the other hand, an asymmetry, similar to that reported in [17], is observed throughout the analyses of the second configuration, where the ring can rotate about its vertical axis by σ , while the offset is kept equal to zero. Below, we summarize the different behaviors observed qualitatively, without showing the corresponding representative figures:

- (1) for $\sigma \leq -61^\circ$: no reconnections are observed;
- (2) for $-60^\circ \leq \sigma \leq -58^\circ$: reconnections are observed;
- (3) for $-57^\circ \leq \sigma \leq -33^\circ$: absorption occurs, similarly to the behavior reported in [18];

- (4) for $-32^\circ \leq \sigma \leq -29^\circ$: a detached disk-like density structure is observed, which subsequently loses consistency;
- (5) for $-28^\circ \leq \sigma \leq -17^\circ$: as the angles approach positive values, the disk persists and eventually transforms into a vortex ring, leading to a reconnection;
- (6) for $-16^\circ \leq \sigma \leq 21^\circ$: reconnection occurs;
- (7) for $\sigma \geq 22^\circ$: no reconnections are observed.

Geometrical and energy investigations regarding the second setting will be reported in the following sections. Based on the results above, we have chosen $\sigma = -75^\circ$, $\sigma = -60^\circ$, $\sigma = -45^\circ$, $\sigma = -30^\circ$, $\sigma = -15^\circ$, $\sigma = 15^\circ$, $\sigma = 30^\circ$, as shown in Fig. 4. We don't consider the base case $\sigma = 0^\circ$, since it coincides with the case $\alpha = 0^\circ$.

A. Behavior of total length and curvature during the reconnection process

For each value of the angle α , we have computed the total length (4), which is reported in Fig. 5. It is clear that, regardless of the initial angle, it increases before the reconnections (indicated by vertical dashed lines) and subsequently decreases, thus generating two peaks corresponding to the two reconnections. This is consistent with the results in [1–3]. Since reconnection events generate sound waves (see Sec. III C), the decrease in the total length is associated to sound emission [12]. On the other hand, we observe that far away from the reconnections the total length has approximately the same value for any α , reflecting the fact that reconnection is a local phenomenon. More precisely, the total length at initial and final times is approximately the same. Furthermore, we observe that the interval between the two reconnections increases with increasing α , implying that the second reconnection occurs more slowly.

The time evolution of the total curvature (4) is presented in Fig. 6. Before the first reconnection, the straight vortex line undergoes a deformation from the beginning, and the signature of this phenomenon is visible in the increase of the curvature of the line (denoted by red squares) as the first reconnection approaches. The curvature of the vortex ring (denoted by green circles) remains close to its initial value $\frac{1}{R} = \frac{1}{8} = 0,125$ until the first reconnection. After this event, the total curvature of the single curve (denoted by blue stars) remains approximately constant, with a value close to the average of curvatures of the two individual curves just before the first reconnection. Finally, after the second reconnection, the total curvatures of the vortex line and ring increase compared to their respective values before the first reconnection, thus possibly reflecting the irreversible nature of vortex reconnections, as pointed out in [3,26,27]. We observe that the total curvature of both the vortex ring and the vortex line presents oscillations after the second reconnection with particular emphasis in the cases $\alpha = 0^\circ$, $\alpha = 15^\circ$, $\alpha = 30^\circ$, and $\alpha = 45^\circ$. These oscillations emerge in the simulation as the vortex ring deforms into a distorted ellipse, with a higher local curvature at its ends, and the line becomes perturbed, suggesting the presence of Kelvin waves (see Sec. III C for further details). We notice that, for $\alpha > 45^\circ$, after the second reconnection the total curvature of the vortex ring increases monotonically with the angle.

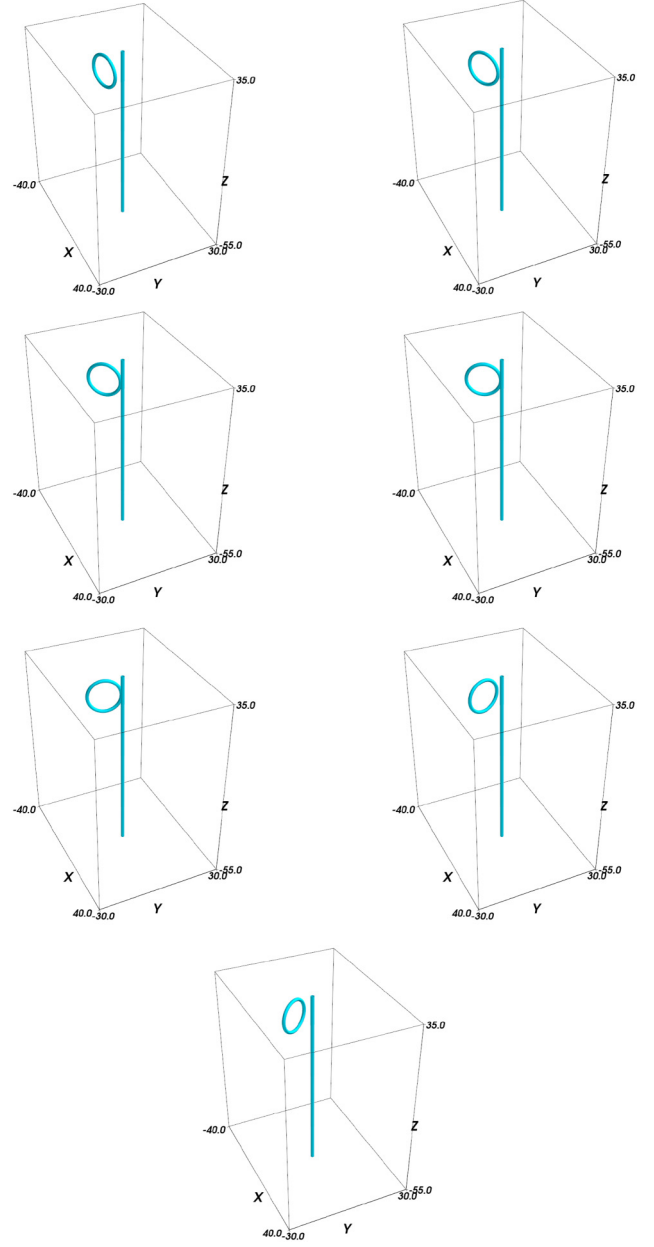


FIG. 4. Configurations of initial conditions with different values of angle σ . From top left to bottom right, $\sigma = -75^\circ$, $\sigma = -60^\circ$, $\sigma = -45^\circ$, $\sigma = -30^\circ$, $\sigma = -15^\circ$, $\sigma = 15^\circ$, and $\sigma = 15^\circ$. Center of the vortex ring: $C_R = [-20 \ 0 \ 15]$. Ring normal coordinates: $\vec{n}_R = [1, \tan \sigma, 0]$. Center of the vortex line: $C_L = [0 \ 0 \ 0]$. Line direction coordinates: $\vec{n}_L = [0 \ 0 \ -1]$. Isosurfaces of $\rho = 0.2$ are plotted to visualize the vortex cores.

Figure 7 shows the total length as a function of time for each value of the angle σ . It is clear that L remains approximately constant for $\sigma = -75^\circ$ and $\sigma = 30^\circ$, which corresponds to the configurations where no reconnection events take place. In contrast, for $\sigma = -60^\circ$, L exhibits two different peaks associated with vortex reconnections, whereas less pronounced double peaks are observed for $\sigma = -15^\circ$ and $\sigma = 15^\circ$ due to the very short time interval separating the two reconnections. Moreover, for these three angles ($\sigma = -60^\circ$, -15° , and 15°), the total length evaluated at the initial

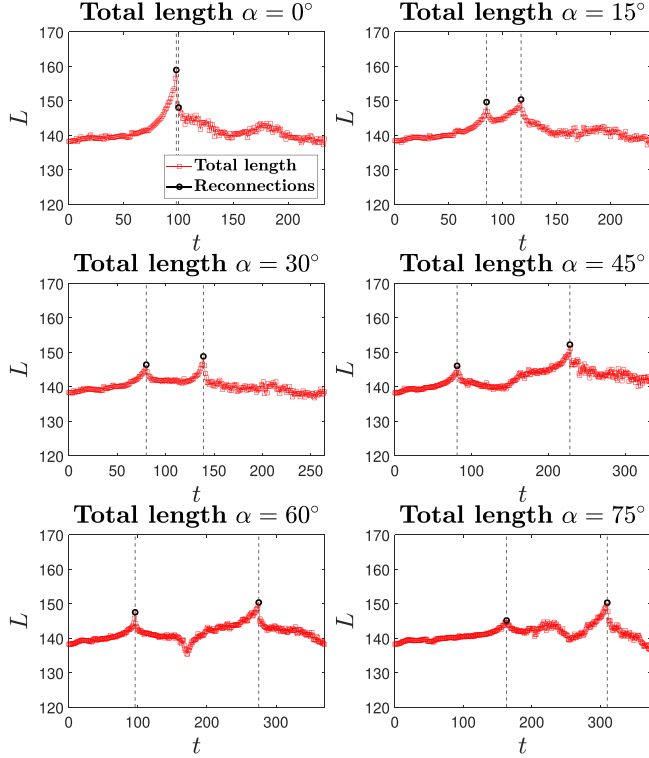


FIG. 5. Total length L (nondimensionalized by the healing length ξ) of nodal lines versus time t for each angle α . Vertical dashed lines indicate reconnection events. The absolute value of the difference between the final $L(T)$ and initial length L_0 for $\alpha = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$ is, respectively, $|L(T) - L_0| = 0.6321, 0.9640, 0.2299, 3.1979, 0.0753, 1.4332$.

and final times shows only a slight variation. Finally, for $\sigma = -45^\circ$ and $\sigma = -30^\circ$, we observe an initial increase in the total length followed by a peak and a subsequent decrease. For these specific values of the angle σ , the vortex ring is absorbed ($\sigma = -45^\circ$), or a disk-like density structure detaches and later loses coherence ($\sigma = -30^\circ$); this behavior suggests a clear correlation between the loss of length (evaluated within the numerical domain) and the absorption phenomenon occurring during the dynamics.

In Fig. 8, we report the time evolution of the total curvature (8) for each value of the angle σ . Similar to the length analysis, different behaviors can be identified for the total curvature depending on the occurrence of reconnection and absorption events. For the cases without reconnection and absorption events, the total curvature of both the vortex ring and the vortex line does not vary significantly. Specifically, the curvature of the line (represented by red squares) is zero at $t = 0$ and subsequently undergoes a minor increase, remaining approximately constant with only minor perturbations due to the small deformation of the line.

On the other hand, for the cases characterized by two reconnections ($\sigma = -60^\circ, -15^\circ, 15^\circ$), the total curvatures of both the vortex ring and the vortex line after the second reconnection increase compared to their values prior to the first reconnection, except for $\sigma = 15^\circ$, where the portions of both the vortex ring and the vortex line involved in the reconnection are much smaller with respect to all the other cases, thus

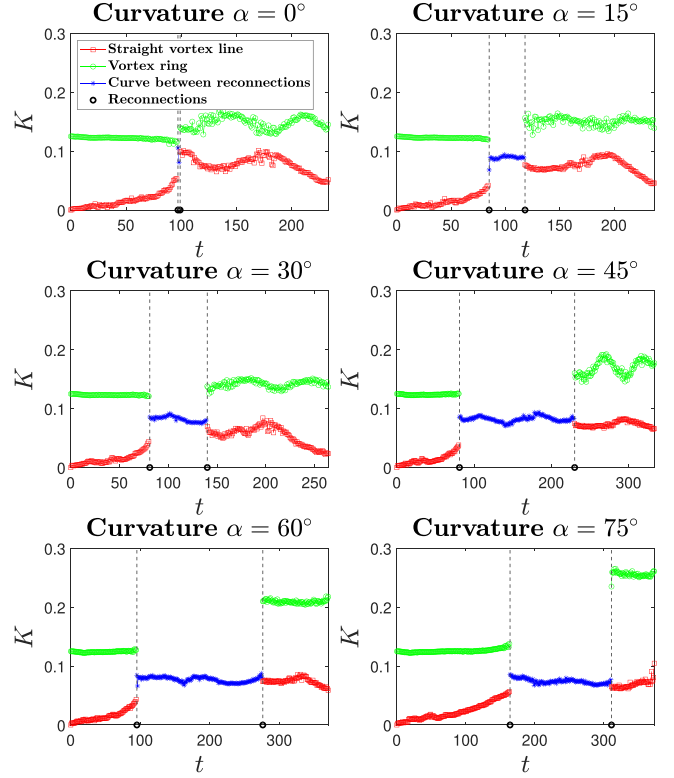


FIG. 6. Total curvature K of nodal lines versus time t for each angle of inclination α of the vortex ring. The green circles represent the total curvature of the vortex ring, the red squares indicate the total curvature of the straight vortex line, and the blue stars indicate the total curvature of the curve between the two reconnection events. Vertical dashed lines denote reconnection events.

leading to lower values of the total curvatures. Moreover, the curvature of the line formed between the reconnection events (denoted by the blue star line) remains constant at a value corresponding to the average curvature of both the vortex line and the vortex ring. We emphasize that, since the time interval between the two reconnections is remarkably short for $\sigma = -15^\circ$ and $\sigma = 15^\circ$, the curvature of this intermediate line is present only for a few time steps. Finally, for $\sigma = -45^\circ$ and $\sigma = -30^\circ$, the total curvature of the vortex ring (denoted by green circles) is initially equal to $1/R = 1/8 = 0.125$ and remains approximately constant until the absorption event, whereas the total curvature of the vortex line starts from zero and increases progressively up to that event. Furthermore, the line formed after the absorption initially exhibits an average value closer to the total curvature of the ring, remaining almost constant thereafter.

B. Behavior of energy components during the reconnection process

In Fig. 9, we report the time evolution of each energy component (2) for different values of α . Since they all change in mean linearly in time, we perform a linear least-squares fit, obtaining the slopes m_{E_q} , m_{E_k} , m_{E_p} , and $m_{E_{int}}$ of the quantum, kinetic, potential, and interaction energy components, respectively. We stress the fact that the energy components are an

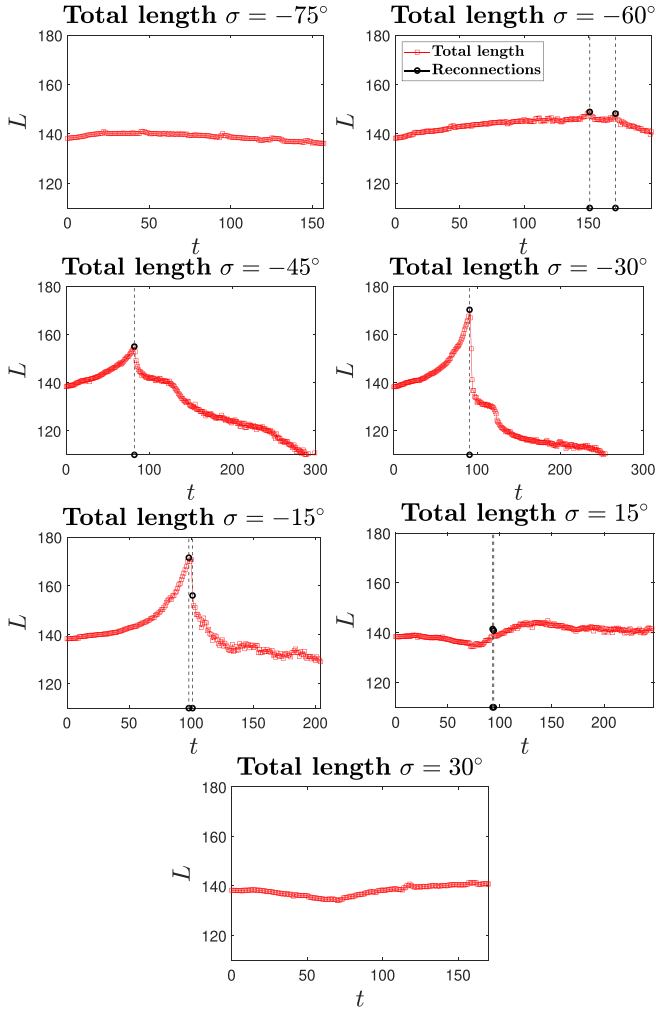


FIG. 7. Total length L (nondimensionalized by the healing length ξ) of nodal lines versus time t for each angle σ . Vertical dashed lines indicate reconnection events. The absolute value of the difference between the final $L(T)$ and initial length L_0 for $\alpha = -75^\circ, -60^\circ, -45^\circ, -30^\circ, -15^\circ, 15^\circ, 30^\circ$ is, respectively, $|L(T) - L_0| = 2.0065, 2.5243, 29.4309, 39.718, 9.3965, 3.2499, 2.5321$.

average over the fluid, and not the components of the energy of the vortex filaments. It is immediately evident that $m_{E_k} + m_{E_q} + m_{E_p} + m_{E_{\text{int}}} = m_{E_{\text{tot}}} = 0$, thus confirming that the energy components compensate each other, being the energy constant throughout the dynamics. To better analyze their behaviors, the linear fits are plotted in a single figure for different values of α (see Fig. 9: quantum energy in blue diamonds, kinetic energy in light-blue squares, potential in green triangles, and internal energy in purple dots). It is clear that

(1) m_{E_k} decreases with increasing angle α , $m_{E_k} > 0$ for $\alpha < 45^\circ$ and $m_{E_k} < 0$ for $\alpha \geq 45^\circ$. This suggests that the kinetic energy E_k increases for $\alpha < 45^\circ$, with a progressively diminishing positive rate of change. For $\alpha \geq 45^\circ$, E_k decreases and the rate of change becomes increasingly negative;

(2) m_{E_q} is approximately constant for all values of α and $m_{E_q} > 0$. This indicates that the quantum energy E_q increases with α , maintaining a uniform rate of change. For $\alpha = 0^\circ$, the outcomes are the same as those obtained in [17];

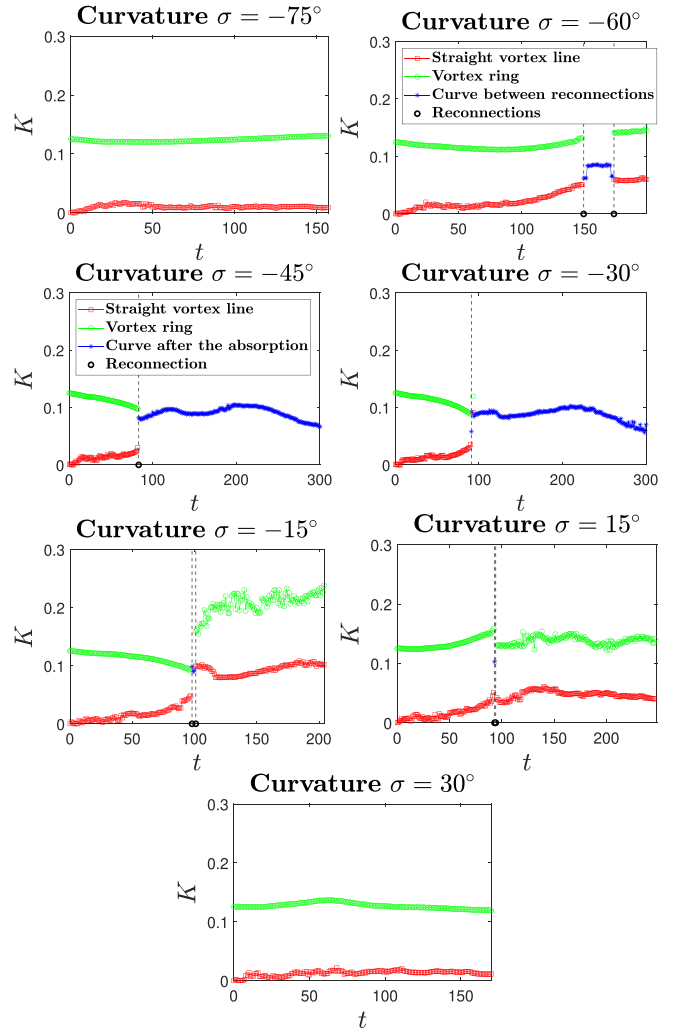


FIG. 8. Total curvature K of nodal lines versus time t for each angle of inclination σ of the vortex ring. The green circles represent the total curvature of the vortex ring, the red squares indicate the total curvature of the straight vortex line, and the blue stars indicate the total curvature of both the curves between the two reconnection events and after the absorption of the vortex ring, respectively. The vertical dashed lines denote reconnection events.

(3) $m_{E_{\text{int}}}$ increases with increasing angle α , $m_{E_{\text{int}}} < 0$ for $\alpha < 45^\circ$ and $m_{E_{\text{int}}} > 0$ for $\alpha \geq 45^\circ$. This means that the internal energy E_{int} decreases for $\alpha < 45^\circ$ with a negative rate of change that progressively diminishes, and it increases for $\alpha \geq 45^\circ$, with an increasing trend characterized by a growing positive slope;

(4) $m_{E_p} = 0$ for every angle α , indicating that the potential energy E_p is constant for each value of α .

Based on these facts, it follows that for each α the kinetic, quantum, and internal energy components play the main role in the redistribution of energy [see Eq. (2)] as the potential component remains constant. Specifically, the kinetic E_k and internal E_{int} energies exhibit opposite behavior, and this happens at $\alpha \approx 45^\circ$. Further checks revealed that this exchange occurs at $\bar{\alpha} = 43^\circ$. It should be noticed that the sum of kinetic and internal energy is compensated by the constant increase

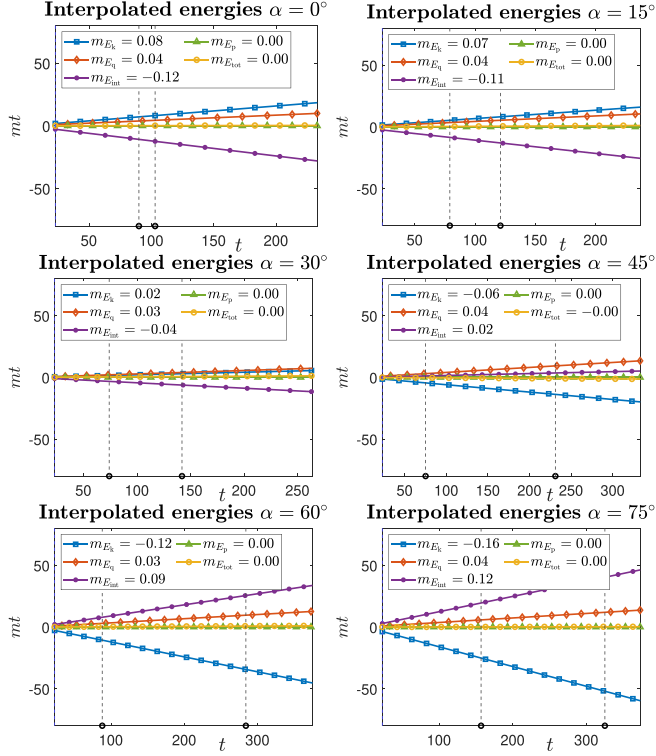


FIG. 9. Linear least-squares fit of energy components (2) for different values of the angle α : kinetic energy (light blue squares), quantum energy (orange diamonds), potential energy (green triangles), internal energy (purple dots), and total energy (yellow circles). The respective angular coefficients m of the fit lines are located in the legends. The plot starts from $t \approx 23$ (the time necessary for the solution of the Eq. (1) to stabilize itself) until the final time T , which is different for each α . Dashed lines indicate reconnection events.

rate of the quantum energy E_q . Indeed, $m_{E_k} + m_{E_{int}} = -m_{E_q}$ for every angle α , up to numerical accuracy.

The linear least-squares fit of the time evolution of each total energy component (2) for all values of σ are reported in Fig. 10. We observe again that $m_{E_k} + m_{E_q} + m_{E_p} + m_{E_{int}} = m_{E_{tot}} = 0$. From the obtained slopes, it is clear that

(1) m_{E_k} decreases with increasing angle σ , for $-75^\circ \leq \sigma \leq -15^\circ$, and it increases with increasing angle σ , for $\sigma \geq 15^\circ$. Moreover,

- (a) $m_{E_k} > 0$, for $-75^\circ \leq \sigma \leq -60^\circ$;
- (b) $m_{E_k} < 0$, for $-45^\circ \leq \sigma \leq -15^\circ$;
- (c) $m_{E_k} > 0$, for $\sigma \geq 15^\circ$.

This suggests that the kinetic energy E_k exhibits two different behaviors: one for $\sigma \leq -15^\circ$ and one for $\sigma \geq 15^\circ$. Specifically, E_k increases with a diminishing positive rate of change for $-75^\circ \leq \sigma \leq -60^\circ$, then, after the absorption of the vortex ring, the kinetic energy decreases with a negative rate of change for $-45^\circ \leq \sigma \leq -15^\circ$. Furthermore, E_k increases for $\sigma \geq 15^\circ$, with a growing rate of change;

(2) $m_{E_q} > 0$ for all values of σ ; however, no dependence of m_{E_q} , and hence of the corresponding E_q , on the angle σ is observed;

(3) $m_{E_{int}}$ increases with increasing angle σ for $-75^\circ \leq \sigma \leq -15^\circ$, and it decreases for $\sigma > 15^\circ$. In particular,

- (a) $m_{E_{int}} < 0$ for $-75^\circ \leq \sigma \leq -60^\circ$;

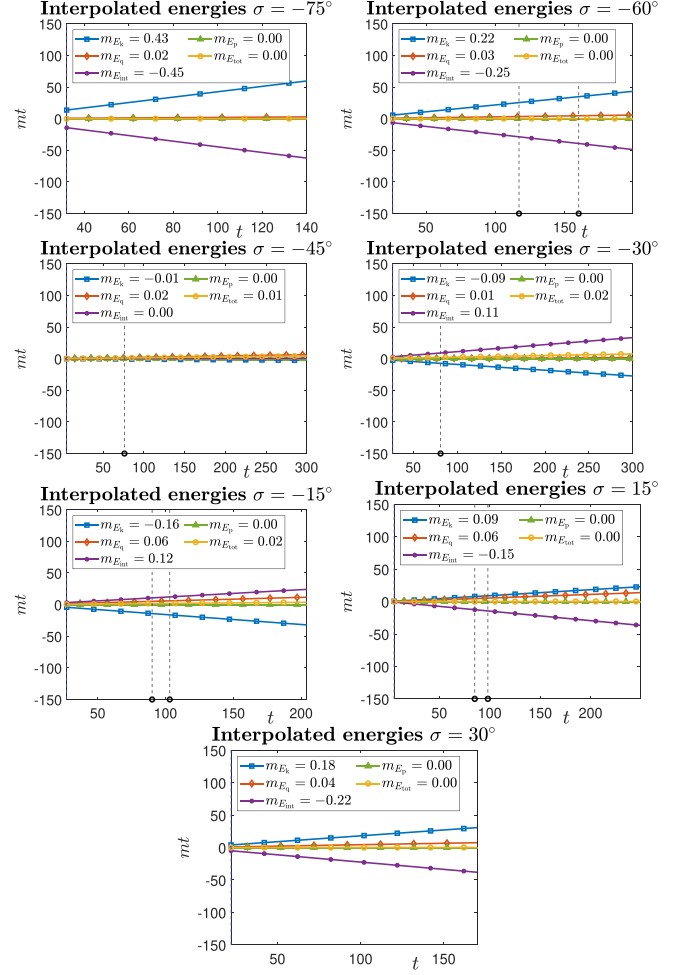


FIG. 10. Linear least-squares fit of energy components (2) for different values of the angle σ : kinetic energy (light blue squares), quantum energy (orange diamonds), potential energy (green triangles), internal energy (purple dots) and total energy (yellow circles). The respective angular coefficients m of the fit lines are located in the legends. The plot starts at a nonzero time, which varies with σ and corresponds to the time required for the solution of Eq. (1) to stabilize. Also, the final time T is different for each value of σ . Dashed lines indicate reconnection events.

- (b) $m_{E_{int}} \leq 0$ for $-45^\circ \leq \sigma \leq -15^\circ$;
- (c) $m_{E_{int}} < 0$ for $\sigma \geq 15^\circ$.

The results indicate that the internal energy E_{int} also exhibits two different behaviors for $\sigma \leq -15^\circ$ and for $\sigma \geq 15^\circ$. Specifically, E_{int} decreases with a diminishing negative rate of change for $-75^\circ \leq \sigma \leq -60^\circ$, then, after the absorption of the vortex ring, the internal energy increases with a growing positive rate of change for $-45^\circ < \sigma \leq -15^\circ$. Furthermore, E_{int} decreases for $\sigma \geq 15^\circ$, with a declining rate of change;

(4) $m_{E_p} = 0$ for every angle σ , indicating that the potential energy E_p is constant also for each value of σ .

C. Behavior of incompressible and compressible contributions to kinetic energy during the reconnection process

For each time step, we first compute the compressible and incompressible contributions to the kinetic energy by

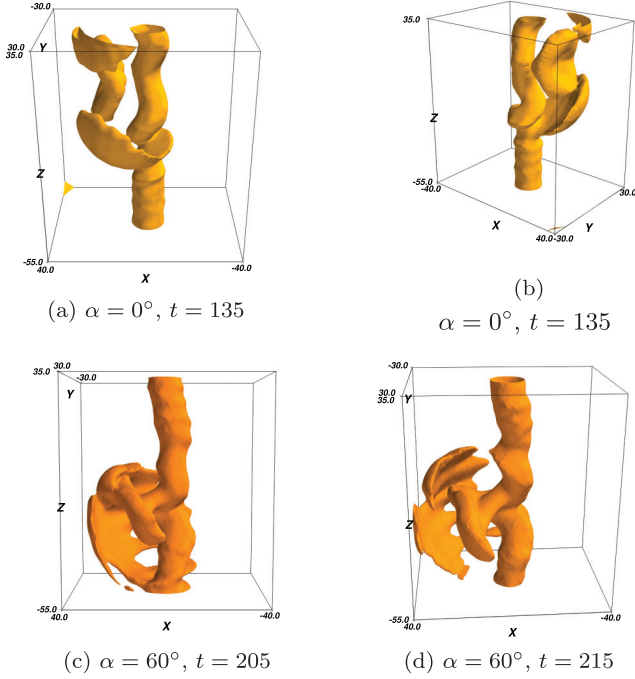


FIG. 11. Emission of the sound pulse during the second reconnection is shown from different perspectives: for $\alpha = 0^\circ$ in panels (a) and (b), and for $\alpha = 60^\circ$ in panels (c) and (d). The isosurfaces are set at $\rho = 0.98$. For the complete simulations including all angle variations, see the Supplemental Material [24].

performing the Helmholtz decomposition outlined in Sec. II C, and then we linearly fit them using the same procedure as in Sec. III B. The evolution of incompressible (red triangles) and of the compressible (black crosses) components for each value of angle α is reported in Fig. 12, together with the slopes $m_{E_k^i}$ and $m_{E_k^c}$, respectively. It is evident that $m_{E_k^i} + m_{E_k^c} = m_{E_k}$, so Eq. (5) is satisfied. For completeness, the fit of the total kinetic energy is also displayed. We observe that

(1) $m_{E_k^i} < 0$ for each α and decreases as α increases; thus E_k^i decreases for each value of α , and the decreasing rate becomes more significant as the angle increases;

(2) $m_{E_k^c} > 0$ for each α , implying an increase in compressible kinetic energy E_k^c , which is related to the emission of sound waves, as we can observe in Fig. 11 where isosurfaces of $\rho = 0.98$ are plotted showing the pulse of sound that is emitted after the reconnection events. However, $m_{E_k^c}$ decreases as the angle increases, indicating that the increase of E_k^c becomes less pronounced as the angle α grows; therefore, the production of sound waves is more intense at $\alpha = 0^\circ$, corresponding to the first configuration shown in Fig. 3.

We also notice that as the angle increases, the fit of the kinetic energy seems to head from the compressible to the incompressible component, suggesting that the more the angle increases, the more the kinetic energy is composed mainly of its incompressible part.

The evolution of the incompressible (red triangles) and of the compressible (black crosses) components of the kinetic energy, for each value of angle σ , is reported in Fig. 13. Also, for the study of the angle σ , it can be observed that $m_{E_k^i} +$

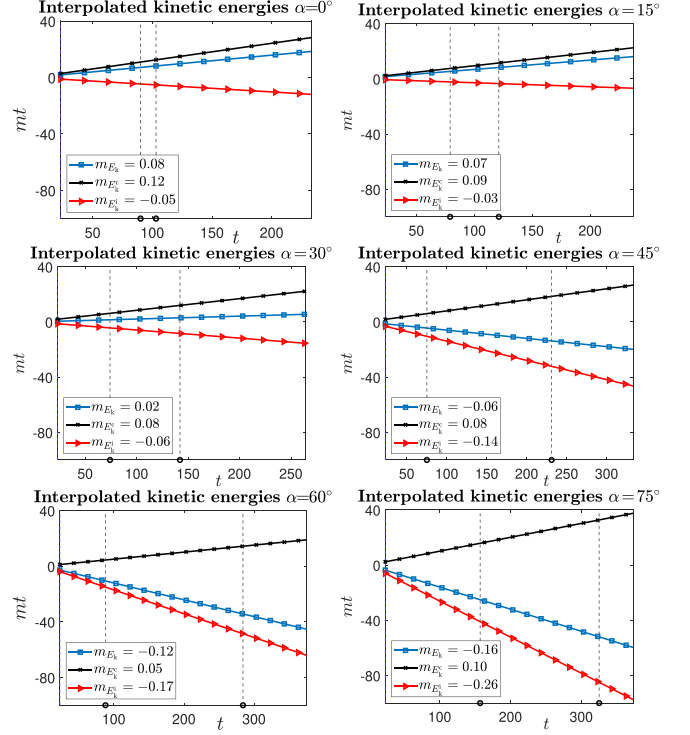


FIG. 12. Linear least-squares fit of kinetic energy decomposition (5) for different values of the angle α : compressible kinetic energy (black crosses), incompressible kinetic energy (red triangles), kinetic energy (light blue squares). The respective angular coefficients m of the fit lines are located in the legends. We recall that the plot starts from $t = 23$ (the time necessary for the solution of the Eq. (1) to stabilize itself) until the final time T , which is different for each α . Dashed lines indicate reconnection events. For $\alpha = 0^\circ$, the kinetic energy decomposition leads to the same results as those obtained in [17].

$m_{E_k^c} = m_{E_k}$ (within numerical accuracy), so Eq. (5) is satisfied. For completeness, the fit of the total kinetic energy is also displayed. From the results, it is clear that

(1) $m_{E_k^i}$ exhibits a trend that follows that of m_{E_k} ; in fact, it decreases for $-75^\circ \leq \sigma \leq -15^\circ$ and increases for $\sigma \geq 15^\circ$. Furthermore,

- (a) $m_{E_k^i} > 0$ for $-75^\circ \leq \sigma \leq -60^\circ$;
- (b) $m_{E_k^i} < 0$ for $-45^\circ \leq \sigma \leq -15^\circ$;
- (c) $m_{E_k^i} > 0$ for $\sigma \geq 15^\circ$.

The results suggest that the incompressible kinetic energy E_k^i increases for $-75^\circ \leq \sigma \leq -60^\circ$ with a decreasing rate, then decreases for $-45^\circ \leq \sigma \leq -15^\circ$ with an increasing negative rate; finally, for $\sigma \geq 15^\circ$, E_k^i increases with a positive increasing rate;

(2) $m_{E_k^c} > 0$ for each σ , and it increases for both $-75^\circ \leq \sigma \leq -15^\circ$ and $\sigma \geq 15^\circ$, indicating that E_k^c increases and that the associated sound wave emission becomes more pronounced as σ grows in these ranges.

IV. CONCLUDING REMARKS

We have studied the reconnection between a vortex ring and a straight vortex line within the framework of superfluids, whose dynamics is governed by the GPE, considering two

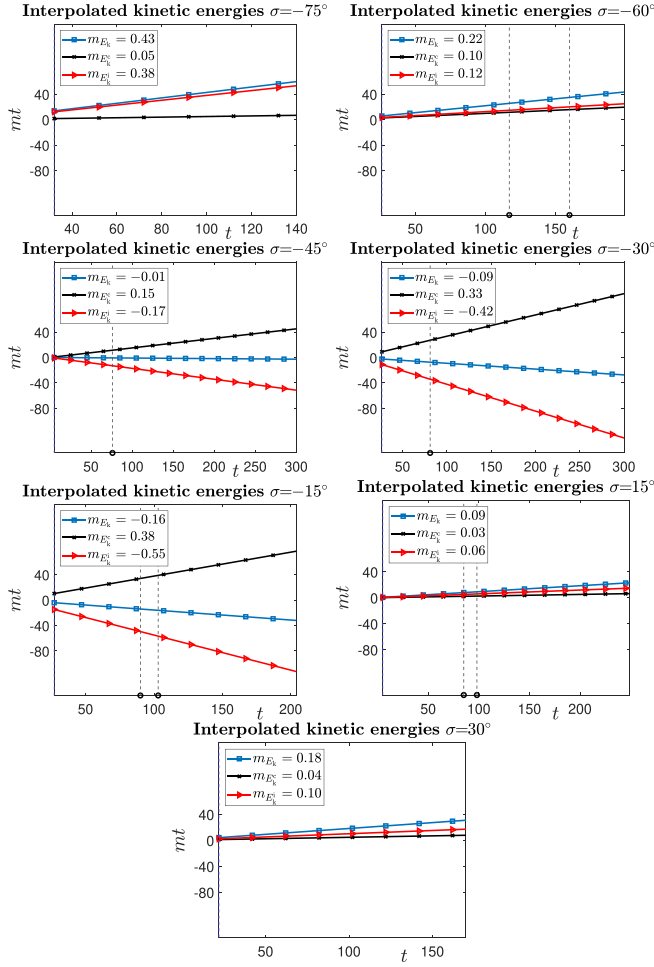


FIG. 13. Linear least-squares fit of kinetic energy decomposition (5) for different values of the angle σ : compressible kinetic energy (black crosses), incompressible kinetic energy (red triangles), kinetic energy (light blue squares). The respective angular coefficients m of the fit lines are located in the legends. We recall that the plot starts from the time necessary for the solution of the Eq. (1) to stabilize itself until the final time T , which is different for each α . Dashed lines indicate reconnection events.

different families of initial configurations: one obtained by varying the angle α between the vortex ring and the horizontal plane, and the other one by changing the angle σ between the vortex ring and the vertical plane xz through the straight vortex and the ring center. As far as the first setting is concerned, we observed that the vortex ring and the straight vortex line first reconnect in a single curve, and then reconnect again, giving rise to a curved vortex line and a perturbed vortex ring. On the other hand, the value of the angle σ significantly affects the time evolution of the second configuration, leading to a reconnection, as in the first case, or to an absorption of the vortex ring by the straight vortex line for $-57^\circ \leq \sigma \leq -33^\circ$.

We have analyzed the behavior of the total length and curvature of the vortex centerlines and the redistribution of energy among its components as the angles α and σ vary. For every angle α , the total length exhibits two peaks corresponding to the reconnections, while the curvatures of the ring and the line increase after the second reconnection with respect to

their values before the first reconnection. From the analysis of the second configuration, it emerges that for the values of σ at which reconnections occur, the total length presents two peaks which are less pronounced with respect to the total length peaks obtained from the analysis of the angle α . Correspondingly, the simulations show that the curve formed between the two reconnections persists only briefly compared with the first configuration, resulting in a much faster transition, with the ring passing through the line without inducing significant distortion. For the angle $\sigma = -45^\circ$, the total length exhibits one peak before decreasing, in agreement with the simulations, which reveal that there is an absorption of the ring by the line. For the values of σ which ensure reconnections, the total curvature of both the vortex ring and the vortex line increases compared with the values before reconnection, whereas for the values of σ that lead to the absorption of the ring, the curvature of the line after absorption remains approximately constant, with a numerical value close to the initial curvature of the vortex ring.

By linearly fitting the time evolution of the different components of energy, we have found the existence of a critical angle $\bar{\alpha}$ at which the coefficients m_{E_k} and $m_{E_{int}}$ (kinetic and internal components, respectively) change signs. In particular, the coefficient m_{E_k} is positive for $\alpha < \bar{\alpha}$ and negative otherwise, whereas the opposite occurs for $m_{E_{int}}$. The slope m_{E_q} of the quantum contribution to the total energy remains consistently positive independently of the angle α ; therefore, the quantum energy grows at a constant rate regardless of the angle and, due to the conservation of total energy, the internal energy must compensate for the trend of the kinetic energy by showing an opposite behavior. The further decomposition of the kinetic energy into its incompressible and compressible components shows that the coefficient $m_{E_k^i}$ of the linear fit of the former is negative and decreases with increasing α , whereas the coefficient $m_{E_k^c}$ of the latter is positive and decreases with α , a clear sign of the generation of sound waves.

The fact that the curvature is associated, via the localized induction approximation (LIA) [6], with the velocity of the nodal lines and thus with their kinetic energy, and that for each angle the total curvature of both the line and the vortex ring after the second reconnection increases with respect to their initial values, allows us to deduce that the loss of incompressible kinetic energy in the fluid is compensated by a corresponding increase in the kinetic energy of the lines themselves.

The decomposition of kinetic energy also reveals that the generation of sound waves is maximal at $\alpha = 0^\circ$. As the angle increases, the growth rate of the compressible kinetic energy progressively decreases. This behavior is further confirmed by the analysis of the isosurfaces of density (see Fig. 11), which shows that the acoustic impulse is weaker as the angle α increases.

On the contrary, for increasing α , the kinetic energy is predominantly composed of the incompressible component, which exhibits a more pronounced decrease with growing α . This leads to a progressive decline in the value of m_{E_k} , further influenced by the attenuation of the growth rate of the compressible component. For $\alpha > \bar{\alpha}$, m_{E_k} becomes negative, indicating a gradual decrease in kinetic energy beyond these angular values. Correspondingly, for $\alpha > \bar{\alpha}$, we have noticed

an increase in the curvature of the ring after the second reconnection. We again deduce that, since curvature is associated with the velocity of the nodal lines, the progressive loss of the fluid's incompressible kinetic energy with increasing angle is offset by a progressive increase in the kinetic energy of the nodal lines.

From the energy analysis derived by varying the angle σ , the fits reveal two different behaviors for negative and positive values of σ . For $\sigma < 0$, we again identify a critical angle $\bar{\sigma} \approx -45^\circ$, beyond which the coefficients m_{E_k} and $m_{E_{int}}$ change sign. More specifically, m_{E_k} is positive for $\sigma < \bar{\sigma}$ and negative otherwise, whereas the opposite holds for $m_{E_{int}}$. Interestingly, this sign reversal occurs at angular values corresponding to the absorption of the vortex ring. For $\sigma > 0$, by contrast, m_{E_k} remains positive and increases, while $m_{E_{int}}$ remains negative and decreases. Unlike the analysis performed for the angle α , the results do not reveal any clear constant behavior or dependence of m_{E_q} on the variation of the angle σ , although the coefficient remains positive for all values of σ .

From the decomposition of the kinetic energy in the case $\sigma \neq 0^\circ$, it emerges that the coefficient $m_{E_k^i}$ of the fit of the incompressible kinetic energy follows the same trend as the kinetic energy. In particular, for $\sigma < 0^\circ$ it exhibits progressively decreasing values as the angle increases and remains positive up to $\bar{\sigma}$, where it changes sign. We observe again that as the angle increases, the curvature of the vortex ring after the second reconnection grows; hence, we again suggest that the progressive loss of the fluid's incompressible kinetic energy with increasing angle is offset by a progressive increase in the kinetic energy of the nodal lines. Conversely, for $\sigma > 0^\circ$, $m_{E_k^i}$ is positive and increases with the angle. Furthermore, the coefficient $m_{E_k^c}$ is positive and increases with the angle for $\sigma < 0^\circ$, indicating that a pronounced sound emission is produced for $\sigma = -15^\circ$, then it takes smaller values and subsequently increases again for $\sigma > 0^\circ$.

It is evident that, for the second configuration where σ is varied, the circulation of the straight vortex line introduces an asymmetry in both the dynamical evolution and the energy decomposition. In particular, two different behaviors can be identified for $\sigma > 0^\circ$ and $\sigma < 0^\circ$. In the latter case, where vortex absorption is also observed, the circulation induces a flow that opposes the motion of the ring. This is reflected in the trend of the fluid kinetic energy, which decreases for negative values of σ , especially for the angles corresponding to stronger vortex interactions. Conversely, for positive values of σ , the fluid kinetic energy increases, as the flow induced by the circulation of the line vortex is aligned with and therefore supports the motion of the ring.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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